

BHABHA UNIVERSITY, BHOPAL

NH-12, Hoshangabad Road, Jatkhedhi, Bhopal-462026

MASTER IN COMPUTER APPLICATION (MCA)

Fifth Semester

Scheme of Examination



DETAILED SYLLABUS (EFFECTIVE FROM JULY 2018)

**Department of Computer Science & Application
BHABHA UNIVERSITY, BHOPAL**

MASTER IN COMPUTER APPLICATION (MCA)

FIFTH SEMESTER

Scheme of Examination

S.No	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem Exam	Tests	Assignment/Quiz	End Sem Practical Viva	Practical Record/Assignment/Quiz/Presentation	
1	MCA51	Cloud computing	3	1	-	4	70	20	10		-	100
2	MCA52	Data warehousing & Mining	3	1	-	4	70	20	10	-	-	100
3	MCA53	Software project management	3	1	-	4	70	20	10	-	-	100
4	MCA54	Elective II	3	1	-	4	70	20	10	-	-	100
5	MCA55	Elective III	3	1	-	4	70	20	10	-	-	100
6	MCA56	Minor project II	-	-	8	8	-	-	-	100	50	150
7	MCA57	Cloud computing Lab	-	-	2	2	-	-	-	30	20	50
		Total	15	5	10	30	350	100	50	130	70	700

L: Lecture - T: Tutorial - P: Practical

Elective II

- A) Distributed System
- B) Java Technology
- C) Network Security

Elective III

- A) Internet and web technology
- B) Modeling & Simulation
- C) Advance DBMS

BHABHA UNIVERSITY, BHOPAL (M.P.)

MCA-51 Cloud Computing

Unit-I

Introduction: Cloud computing ,Vision and Characteristics of cloud computing as per NIST , Cloud computing reference model ,Cloud computing environments, Cloud services requirements, Cloud and dynamic infrastructure, Cloud Adoption and rudiments .Overview of cloud applications: ECG Analysis in the cloud, Protein structure prediction, Gene Expression Data Analysis ,Satellite Image Processing ,CRM and ERP ,Social networking .

Unit-II

Architecture: Cloud Reference Model, Types of Clouds, Cloud Interoperability & Standards, Scalability and Fault Tolerance, Cloud Solutions: Cloud Ecosystem, Cloud Business Process Management, Cloud Service Management. Cloud Offerings: Cloud Analytics, Testing Under Control, Virtual Desktop Infrastructure.

Unit –III

Cloud Management & Virtualization Technology: Resiliency, Provisioning, Asset management, Concept of Map reduce , Cloud Governance, High Availability and Disaster Recovery. Virtualization: Fundamental concepts of compute ,storage, networking, desktop and application virtualization .Virtualization benefits, server virtualization, Block and file level storage virtualization Hypervisor management software, Infrastructure Requirements , Virtual LAN(VLAN) and Virtual SAN(VSAN) and their benefits .

Unit-IV

Cloud Information security fundamentals, Cloud security services, Design principles, Secure Cloud Software Requirements, Policy Implementation, Cloud Computing Security Challenges, Virtualization security Management, Cloud Computing Security Architecture .

Unit-V

Market Based Management of Clouds , Federated Clouds/Inter Cloud: Characterization & Definition ,Cloud Federation Stack , Third Party Cloud Services . Case study: Google App Engine, Microsoft Azure , Hadoop , Amazon , Aneka

Recommended Text:

1. Velte, “Cloud Computing- A Practical Approach” ,TMH Pub
2. Kumar Saurabh, “Cloud Computing” , Wiley Pub
3. Krutz , Vines, “Cloud Security “ , Wiley Pub

MCA-52 Data Warehousing and Mining

UNIT - I

Data warehousing Components –Building a Data warehouse –Mapping the Data Warehouse to a Multiprocessor Architecture , DBMS Schemas for Decision Support – Data Extraction, Cleanup, and Transformation Tools ,Metadata ,reporting ,Query tools and Applications –Online Analytical Processing (OLAP) – OLAP and Multidimensional Data Analysis.

UNIT - II

Data Mining: - Data Mining Functionalities – Data Preprocessing – Data Cleaning – Data Integration and Transformation – Data Reduction – Data Discretization and Concept Hierarchy Generation .Association Rule Mining: - Efficient and Scalable Frequent Item set Mining Methods –~~41~~Mining Various Kinds of Association Rules – Association Mining to Correlation Analysis– Constraint-Based Association Mining.

UNIT - III

Classification and Prediction: - Issues Regarding Classification and Prediction – Classification by Decision Tree Introduction – Bayesian Classification – Rule Based Classification – Classification by Back propagation– Support Vector Machines – Associative Classification – Lazy Learners – Other Classification Methods –Prediction – Accuracy and Error Measures – Evaluating the Accuracy of a Classifier or Predictor – Ensemble Methods – Model Section.

UNIT - IV

Cluster Analysis: - Types of Data in Cluster Analysis ,A Categorization of Major Clustering Methods ,Partitioning Methods , Hierarchical methods ,Density-Based Methods ,Grid-Based Methods , Model-Based Clustering Methods , Clustering High-Dimensional Data ,Constraint-Based Cluster Analysis ,Outlier Analysis.

UNIT - V

Mining Object, Spatial, Multimedia, Text and Web Data: Multi dimensional Analysis and Descriptive Mining of Complex Data Objects , Spatial Data Mining , Multimedia Data Mining ,Text Mining , Mining the World Wide Web.

REFERENCES

1. Jiawei Han and Micheline Kamber “Data Mining Concepts and Techniques” Second Edition,
2. Elsevier, Reprinted 2008.
3. Alex Berson and Stephen J. Smith “Data Warehousing, Data Mining & OLAP”, Tata McGraw – HillEdition, Tenth Reprint 2007.

MCA-53 Software project management

UNIT -I

System concepts: Introduction of system concept, characteristics of system, elements of system, The System Development Life Cycle, The Role of System Analyst. Introduction system planning & initial investigation, various information gathering tools feasibility study conretions & structures tools of system analysis, various methods of process design, form design methodologies, introduction to information system testing, quality assurance security & diastruct computer various (deleting recovery)

UNIT -II

Software Process, Product and Project :

The Product: Software, Software Myths, The process: Software Engineering: A Layered Technology, Software Process Models, The Linear Sequential Model, The Prototyping Model, The RAD Model, Evolutionary Software Process Models, Component – Based Development, Fourth Generation Techniques, Software process and Project Metrics: Software measurement

UNIT-III

Software Project Planning and Design: Software Project Planning: Project planning objectives, Decomposition Techniques, Empirical estimation models, The Make/Buy Decision., Risk analysis.

Software Design: Design Principles, Cohesion & Coupling, Design notation and specification, structure design methodology.

UNIT-IV

Software Quality Assurance and Testing: Software Quality Assurance : Quality Concepts, The Quality Movement, Software Quality Assurance, Software Reviews, Formal Technical Reviews, Formal Approaches to SQA, Statistical Software Quality Assurance, Software Reliability, Mistake Proofing for Software, Introduction to ISO standard. Testing Strategies: A strategic approach of software testing strategic issues, unit testing, integration testing, validation testing, system testing, the art of debugging. OOA, OOD.

UNIT-V

Advanced Topics: MIS & DSS :Introduction to MIS, long range planning, development and implementation of an MIS, applications of MIS in manufacturing sector and in service sector.

Decision Support System concepts, types of DSS. Object Oriented Software Engineering: Object Oriented Concepts, Identifying the Elements of an Object Model, Management of Object Oriented Software Projects. CASE tools, Re-engineering

BOOKS

1. Richard Fairley : Software Engineering Concepts (TMH)
2. Pankaj Jalote “Software Engg” Narosa Publications.
2. Ian Sommerville : Software Engineering 6/e (Addison-Wesley)
3. Richard Fairley : Software Engineering Concepts (TMH)
4. Elis Awad, "System Analysis & Design", Galgotia publications
5. W.S. Jawadekar: Management Information Systems, TMH Publication, India

MCA-54 Elective II(A)
Distributed System

UNIT - I

COMMUNICATION IN DISTRIBUTED ENVIRONMENT :Introduction – Various Paradigms in Distributed Applications – Remote Procedure Call – Remote Object Invocation – Message-Oriented Communication – Unicasting , Multicasting and Broadcasting – Group Communication.

UNIT - II

DISTRIBUTED OPERATING SYSTEMS: Issues in Distributed Operating System – Threads in Distributed Systems – Clock Synchronization – Causal Ordering – Global States – Election Algorithms – Distributed Mutual Exclusion – Distributed Transactions – Distributed Deadlock – Agreement Protocols.

UNIT - III

DISTRIBUTED RESOURCE MANAGEMENT :Distributed Shared Memory – Data-Centric Consistency Models – Client-Centric Consistency Models – Ivy – Munin – Distributed Scheduling – Distributed File Systems – Sun NFS.

UNIT - IV

FAULT TOLERANCE AND CONSENSUS : Introduction to Fault Tolerance – Distributed Commit Protocols – Byzantine Fault Tolerance – Impossibilities in Fault Tolerance.

UNIT - V

CASE STUDIES :Distributed Object-Based System – CORBA – COM+ – Distributed Coordination-Based System – JINI.

REFERENCES:

1. A.S.Tanenbaum , M.Van Steen, “Distributed Systems”, Pearson Education, 2004
2. Hagit Attiya and Jennifer Welch, “Distributed Computing: Fundamentals, Simulations and Advanced Topics”, Wiley, 2004.
3. George Coulouris, Jean Dollimore, Tim Kindberg, “Distributed Systems Concepts and Design”, Third Edition, Pearson Education Asia, 2002.

MCA-54 Elective II(B)
Java Technology

UNIT I

OVERVIEW AND FUNDAMENTALS: Object Oriented Programming concepts , Encapsulation ,Programming Elements Program Structure ,Enumeration Types , Functions and Pointers ,Function Invocation – Overloading Functions – Scope and Storage Class – Pointer Types –Arrays and Pointers, Call-by-Reference ,Strings and exception handling

UNIT II

IMPLEMENTING ABSTRACT DATA TYPES AND ENCAPSULATION: Aggregate Type struct – Structure Pointer Operators – Unions – Bit Fields – Data Handling and Member Functions – Classes – Constructors and Destructors , Static Member – this Pointer – reference semantics – implementation of simple ADTs.

UNIT III

POLYMORPHISM: ADT Conversions – Overloading – Overloading Operators – Unary Operator. Overloading , Binary Operator Overloading , Function Selection , Pointer Operators, Visitation ,

UNIT IV

TEMPLATES: Template Class – Function Templates – Class Templates – Parameterizing – STL –Algorithms – Function Adaptors

UNIT V

INHERITANCE: Derived Class – Typing Conversions and Visibility – Code Reuse – Virtual Functions ,Templates and Inheritance – Run-Time Type Identifications – Exceptions – Handlers –Standard Exceptions.

REFERENCES:

1. Bhavé , “ Object Oriented Programming With C++”, Pearson Education , 2004.
2. Stanley B.Lippman, Josee Lajoie, “C++ Primer”, Pearson Education, Third Edition, 2004.
3. Kamthane,” Object Oriented Programming with ANSI and Turbo C++”, Person Education, 2002.

MCA-54 Elective II(C)
Network Security

UNIT-I

Introduction: Computer Network, Layered Network Architecture-Review of ISO-OSI Model, Transmission Fundamentals-, Communication Media-Conductive Metal (Wired Cable), Optical Fiber links, Wireless Communication-Radio links, Satellite Links, Communication Services & Devices, Telephone System., Integrated Service Digital Network (ISDN)., Cellular Phone., ATM, Modulation & Demodulation-, Digital to Analog Conversion-Frequency Modulation (FM), Amplitude Modulation (AM), Phase Modulation (PM)., Analog to Digital Conversion-Pulse Amplitude Modulation(PAM), Pulse Code Modulation (PCM), Differential Pulse Code Modulation, (DPCM)., Modem & Modem Types, Multiplexing-, Frequency Division Multiplexing (FDM)., Time Division Multiplexing (TDM), Statistical Time Division Multiplexing (STDM)., Contention Protocol-, Stop-Go-Access Protocol, Aloha Protocol- Pure aloha & Slotted aloha, Carrier sense multiple access with collision detection (CSMA/CD)

UNIT-II

Data Security and Integrity: Parity Checking Code, Cyclic redundancy checks (CRC), Hemming Code, Protocol Concepts –, Basic flow control, Sliding window protocol-Go-Back-N protocol and selective repeat protocol, Protocol correctness- Finite state machine

UNIT-III

Local Area Network: Ethernet : 802.3 IEEE standard, Token Ring : 802.5 IEEE standard, Token Bus : 802.4 IEEE standard, FDDI Protocol, DQDB Protocol, Inter Networking, Layer 1 connections- Repeater, Hubs, Layer 2 connections- Bridges, Switches, Layer 3 connections- Routers, Gateways.

UNIT-IV

Wide Area Network: Introduction, Network routing, Routing Tables, Types of routing, Dijkstra's Algorithm, Bellman-Ford Algorithm, Link state routing, Open shortest path first, Flooding, Broadcasting, Multicasting, Congestion & Dead Lock, Internet Protocols, Overview of TCP/IP, Transport protocols, Elements of Transport Protocol, Transmission control protocol (TCP), User data-gram protocol (UDP).

UNIT-V

Network Security, Virtual Terminal Protocol, Overview of DNS, SNMP, email, WWW, Multimedia.

BOOKS

1. A.S.Tanenbaum, "Computer Network", 4th addition, PHI
2. Forouzan "Data Communication and Networking 3ed", TMH

MCA-55 Elective III(A)
Internet and web technology

UNIT-I

Introduction to the Internet including E-mail, World Wide Web (WWW), and how to perform basic research to address company/business needs. Accessing and navigating the Internet, covering concepts, terms, tools, and services such as Internet Service provider (ISP's), newsgroups, and chat rooms. Browsers and search tools.

UNIT -II

BASIC INTERNET CONCEPTS : Connecting to the Internet – Domain Name System - Exchanging Email– Sending and Receiving Files - Fighting Spam, Sorting Mail and avoiding e-mail viruses – Chatting and Conferencing on the Internet – Online Chatting - Messaging – Usenet Newsgroup ,Internet Relay chat (IRC) – Instant Messaging - Voice and Video Conferencing.

UNIT III

WORLD WIDE WEB : Overview – Web Security, Privacy, and site-blocking – Audio and Video on the web ,Creating and Maintaining the Web – Web site creation concepts – Web Page Editors ,Optimizing Web Graphics – Web Audio Files – Forms, Interactivity, and Database- Driven Web sites – File Transfer and downloading – FTP – Peer to Peer – Downloading and Installing software.

UNIT IV

Static and dynamic web pages, tiers, plug-ins, frames and forms. Exposure to Markup languages, HTML, DHTML, XML etc. CGI, Applets & Serve-lets, JSP & JAVA Beans, active X control, ASP cookies creating and reading cookies, Web services, Web application architectures, Browsers, Search engines.

UNIT-V

Advanced technologies for e-commerce: Introduction to mobile agents. WAP: the enabling technology : The WAP model, WAP Architecture, Benefit of WAP to e-commerce. Web Security, Encryption Schemes, secure Web documents, Digital signatures and firewalls.

References:

1. Uttam K: Web Technologies, Oxford University Press.
2. Henry Chan, Raymond Lee, Tharam Dillon , E-Commerce Fundamental and Applications, Willey Publication.
3. Minoli & Minoli, Web Commerce Technology Hand Book, TMH
4. Satyanarayana, E-Government, PH

MCA-55 Elective III(B) Modeling & Simulation

UNIT-I

Introduction: Models, Models types, principles used in modelling, system studies, interacting subsystems and example, simulation definition, examples, steps in computer simulation, advantages and disadvantages of simulation, simulation study, classification of simulation languages.

UNIT-II

System Simulation : Techniques of simulation, monte carlo method, comparison of simulation and analytical methods, numerical computation techniques for continuous and discrete models, distributed leg models, cobweb models.

UNIT-III

Continuous system simulation : Continuous system models, differential equation, analog computer analog methods, digital analog simulators.

System Dynamics : Historical background, exponential, Growth and decay models, modified Exponential growth models, logistic curves and generalization of growth models, system dynamics Diagrams.

UNIT-IV

Probability concepts in simulation : Stochastic variables, discrete and continuous probability function, continuous uniform distributed and computer generation of random numbers, uniform random number generator, non uniform continuously distributed random numbers, rejection method.

Discrete system simulation : Discrete events, representation of time, generation of arrival patterns, simulation of telephone system, delayed calls, simulation programming tasks, gathering statistics, discrete simulation languages.

UNIT-V

Object Oriented approach in simulation, simulation in C++, Introduction to GPSS, general description, action times, choice of paths, simulation of a manufacturing shop, facilities and storage, program control statements, priorities and parameters, numerical attributes, functions, simulation of a supermarket transfer models, GPSS model applied to any application, simulation programming techniques like entry types.

BOOKS

1. Narsingh Deo "System Simulation with Digital Computers", PHI.
2. G.Gordan "System Simulation" , 2nd Ed, 2002 PHI.
3. T.A. Payer "Introduction to Simulation", McGraw Hill.
4. W.A. Spriet "Computer Oriented Modeling and Simulation".

MCA-55 Elective III(C) Advanced Database Management System

UNIT-I

Object Oriented and Object Relational Databases: Modeling Complex Data Semantics, Specialization, Generalization, Aggregation and Association, Objects, Object Identity and its implementation, Clustering, Equality and Object Reference, Architecture of Object Oriented and Object Relational databases, Persistent Programming Languages, Cache Coherence. Case Studies: Gemstone, O2, Object Store, SQL3, Oracle xxi, DB2.

UNIT-II

Parallel and Distributed Databases: Parallel architectures, shared nothing/shared disk/shared memory based architectures, Data partitioning, Intra-operator parallelism, pipelining. Distributed Data Storage – Fragmentation & Replication, Location and Fragment Transparency Distributed Query Processing and Optimization, Distributed Transaction Modeling and concurrency Control, Distributed Deadlock, Commit Protocols, Design of Parallel Databases, and Parallel Query Evaluation.

UNIT-III

Advanced Transaction Processing: Advanced transaction models: Save points, Nested and Multilevel Transactions, Compensating Transactions and Saga, Long Duration Transactions, Weak Levels of Consistency, Transaction Work Flows, Transaction Processing Monitors, Shared disk systems.

UNIT-IV

Active Database and Real Time Databases: Triggers in SQL, Event Constraint and Action: ECA Rules, Query Processing and Concurrency Control, Recursive query processing, Compensation and Databases Recovery, multi-level recovery.

UNIT-V

Image and Multimedia Databases: Modeling and Storage of Image and Multimedia Data, Data Structures – R-tree, k-d tree, Quad trees, Content Based Retrieval: Color Histograms, Textures, etc., Image Features, Spatial and Topological Relationships, Multimedia Data Formats, Video Data Model, Audio & Handwritten Data, Geographic Information Systems (GIS). Accessing Databases through WEB, WEB Servers, XML Databases.

BOOKS

1. Elmagarmid.A.K. “Database transaction models for advanced applications”, Morgan Kaufman.
2. Transaction Processing, Concepts and Techniques, J. Gray and A. Reuter, Morgan Kauffman..
3. D. Maier, “The Theory of Relational Databases”, 1993, Computer Science Press, Rockville, Maryland